



Polynomial

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Let $P(x) = x^3 + 5x^2 + x - 15$.

(i) Show that $P(-3) = 0$.

$$\begin{aligned} \text{(i)} \quad P(-3) &= (-3)^3 + 5(-3)^2 + (-3) - 15 \\ &= -27 + 45 - 3 - 15 \\ &= 0 \end{aligned}$$

1

(ii) Hence, factorise the polynomial $P(x)$ as $A(x)B(x)$, where $B(x)$ is a quadratic polynomial.

2

(ii) Since $P(-3) = 0$, $(x + 3)$ is a factor of $P(x)$

$$\begin{array}{r} x^2 + 2x - 5 \\ x+3 \overline{) x^3 + 5x^2 + x - 15} \\ \underline{x^3 + 3x^2} \\ 2x^2 + x \\ \underline{2x^2 + 6x} \\ -5x - 15 \\ \underline{-5x - 15} \\ 0 \end{array} \quad \therefore P(x) = (x+3)(x^2 + 2x - 5)$$

2 (i) Given that $(x - 3)$ and $(x + 2)$ are factors of $f(x) = x^3 + x^2 + px + q$, find the values of p and q .

2

$$\begin{array}{ll} f(3) = 0 & f(-2) = 0 \\ 27 + 9 + 3p + q = 0 & -8 + 4 - 2p + q = 0 \\ 3p + q = -36 & 2p - q = -4 \end{array}$$

Solve simultaneously,

(ii) Factorise $f(x)$ completely.

2

$$\begin{array}{ll} 5p = -40 & q = -12 \\ p = -8 & \end{array}$$

Therefore,

$$f(x) = (x-3)(x+2)^2$$

$$\begin{array}{r} x+2 \\ x^2 - x - 6 \overline{) x^3 + x^2 - 8x - 12} \\ \underline{x^3 - x^2 - 6x} \\ 2x^2 - 2x - 12 \\ \underline{2x^2 - 2x - 12} \\ 0 \end{array}$$

(iii) For what values of x is $f(x) < 0$? Give your answer using interval notation. $x \in (-\infty, -2) \cup (-2, 3)$

1

3 The polynomial equation $P(x) = ax^3 + 12x^2 + cx + d$ has roots α, β and γ , where a, c and d are real numbers.

(i) Given that $\alpha + \beta + \gamma = 6$ and $\alpha^2 + \beta^2 + \gamma^2 = 32$, find the values of a and c .

2

$$\begin{aligned} \alpha + \beta + \gamma &= -\frac{12}{a} & (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) &= \alpha^2 + \beta^2 + \gamma^2 \\ 6 &= -\frac{12}{a} & 36 - 2\left(\frac{c}{-2}\right) &= 32 \\ a &= -2 & c &= -4 \end{aligned}$$

(ii) Given that the polynomial can be expressed as $P(x) = (x + 2)Q(x)$ where $Q(x)$ is a polynomial, find the value of d .

1

$$\begin{aligned} P(-2) &= 0 \\ -8(-2) + 48 - 2(-4) + d &= 0 \\ \therefore d &= -72 \end{aligned}$$

4 The polynomial $P(x) = x^4 - x^3 - 11x^2 + 9x + 18$ has

four distinct zeroes. Find them.

3

$$\begin{aligned} P(0) &= 18 \neq 0 \\ P(1) &= 1 - 1 - 11 + 9 + 18 \neq 0 \\ P(-1) &= 1 + 1 - 11 - 9 + 18 = 0 \quad \therefore -1 \text{ is a zero} \\ P(2) &= 16 - 8 - 44 + 18 + 18 = 0 \quad \therefore 2 \text{ is a zero.} \\ \text{Let other zeros be } \alpha, \beta. \\ \text{By sum of zeros, } \alpha + \beta + (-1) + 2 &= 1 \\ \therefore \alpha + \beta &= 0 \quad \dots (1) \\ \text{By product of zeros, } \alpha\beta(-1)(2) &= 18 \\ \therefore \alpha\beta &= -9 \quad \dots (2) \\ \text{By inspection, or otherwise, } \alpha &= 3, \beta = -3 \text{ (or reverse!)} \\ \therefore \text{The zeros are } &-1, 2, 3, -3 \end{aligned}$$

5 α, β and γ are the roots of the polynomial $P(x) = 2x^3 - 4x^2 + x + 5$. Calculate:

(i) $3\alpha + 3\beta + 3\gamma - 4\alpha\beta\gamma$ 2

$$\begin{aligned} & \text{i) } 3\alpha + 3\beta + 3\gamma - 4\alpha\beta\gamma \\ & = 3(\alpha + \beta + \gamma) - 4(\alpha\beta\gamma) \\ & = 3\left(-\frac{-1}{2}\right) - 4\left(-\frac{5}{2}\right) \\ & = 6 + 10 \\ & = 16 \end{aligned}$$

(ii) $\alpha^2 + \beta^2 + \gamma^2$ 2

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(-\frac{-1}{2}\right)^2 - 2\left(\frac{1}{2}\right) = 4 - 1 = 3 \end{aligned}$$

6 Determine the values of a and b so that $x^4 - 3x^3 + ax^2 + bx - 1$ is divisible by $(x - 1)^2$. 3

$$\begin{aligned} \text{d) } P(x) &= x^4 - 3x^3 + ax^2 + bx - 1 \\ P'(x) &= 4x^3 - 9x^2 + 2ax + b \\ \text{if divisible by } (x-1)^2 \quad P(1) &= 0 \text{ and } P'(1) = 0 \\ \text{i.e. } P(1) &= 1 - 3 + a + b - 1 = 0 \\ & \quad a + b = 3 \dots \text{①} \\ P'(1) &= 4 - 9 + 2a + b = 0 \\ & \quad 2a + b = 5 \dots \text{②} \\ \text{②} - \text{①} \quad a &= 2 \\ \text{but } a &= 2 \text{ into ①} \quad b &= 1 \quad \therefore a = 2 \quad b = 1 \end{aligned}$$

7 Consider the polynomial $P(x) = x^3 - 2px + q$, where p and q are constants and $p \neq 0$. It is given that α , β and $\alpha + \beta$ are the roots of the equation $P(x) = 0$. 1

(i) Prove that $\alpha = -\beta$.

$$\begin{aligned} a &= 1, b = 0, c = -2p, d = q \\ \text{Taking sum of the roots, one at a time:} \\ \alpha + \beta + (\alpha + \beta) &= -\frac{b}{a} \\ 2(\alpha + \beta) &= 0 \\ \alpha + \beta &= 0 \\ \therefore \alpha + \beta &= 0 \\ \alpha &= -\beta \end{aligned}$$

(ii) Hence, or otherwise, find all the roots of $P(x)$ in terms of p . 3

$$\begin{aligned} \text{Taking sum of the roots, three at a time OR taking product of the roots:} \\ \alpha\beta(\alpha + \beta) &= -q \\ q &= 0 \text{ as } \alpha + \beta = 0 \\ P(x) &= x^3 - 2px + q = x^3 - 2px = x(x^2 - 2p) \\ x(x^2 - 2p) &= 0 \\ x = 0 \text{ or } x^2 - 2p &= 0 \\ x^2 - 2p &= 0 \\ x^2 &= 2p \\ x &= \pm\sqrt{2p} \\ \text{Therefore, the roots are } x &= 0, \pm\sqrt{2p}. \end{aligned}$$

8 Let $P(x) = x^3 + ax^2 + bx + 5$ where a and b are real numbers. Find the values of a and b given that $(x - 1)^2$ is a factor of $P(x)$.

$$\begin{aligned} P(x) &= x^3 + ax^2 + bx + 5 \\ P(1) &= 1 + a + b + 5 = 0 \\ a + b &= -6 \dots \text{①} \\ P'(x) &= 3x^2 + 2ax + b \\ P'(1) &= 3 + 2a + b = 0 \\ 2a + b &= -3 \dots \text{②} \\ \text{②} - \text{①} \quad a &= 3 \\ b &= -9 \quad \checkmark \end{aligned}$$

9 Solve the equation $x^3 + 2x^2 - 5x - 6 = 0$ given that one of its roots is equal to the sum of the other two roots. 2

$$\begin{aligned} x^3 + 2x^2 - 5x - 6 &= 0 \\ \text{Let } \alpha, \beta \text{ and } \alpha + \beta &\text{ be the roots.} \\ \alpha + \beta + \alpha + \beta &= -\frac{2}{1} \\ 2(\alpha + \beta) &= -2 \\ \alpha + \beta &= -1 \\ \alpha &= -1 - \beta \quad \text{--- ①} \\ \alpha\beta(\alpha + \beta) &= -\frac{-6}{1} \\ \alpha\beta(\alpha + \beta) &= 6 \quad \text{--- ②} \\ \text{① into ②} \\ \beta(-1-\beta)(-1-\beta) &= 6 \\ (-\beta - \beta^2)(-1) &= 6 \\ \beta^2 + \beta - 6 &= 0 \\ (\beta + 3)(\beta - 2) &= 0 \\ \beta &= -3, \beta = 2 \quad \text{--- ③} \\ \text{③ into ①} \\ \alpha &= -1 - (-3) \quad \alpha = -1 - 2 \\ \alpha &= 2 \quad \alpha = -3 \end{aligned}$$

The roots are 2, -3 and -1.



Polynomial

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 When the polynomial $P(x)$ is divided by $x^2 - 16$ the remainder is $3x - 1$. What is the remainder when $P(x)$ is divided by $x - 4$? 2

$$\begin{aligned}P(x) &= (x^2 - 16)Q(x) + (3x - 1) \\ \text{remainder} &= P(4) \\ &= 0 \cdot Q(4) + 11 \\ &= \boxed{11}\end{aligned}$$

- 2 Consider the equation $x^2 + 2x + 3 = 0$. What is the sum of the reciprocal of the roots of this equation? 1

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-2}{3}$$

- 3 If $x = \alpha$ is a double root of a polynomial $P(x)$, show that $x = \alpha$ is also a root of $P'(x)$. 2

Hint: Let $P(x) = (x - \alpha)^2 Q(x)$.

$$\begin{aligned}P(x) &= (x - \alpha)^2 Q(x) \\ P'(x) &= 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x) \\ P'(\alpha) &= 0 + 0 = 0 \\ \therefore x = \alpha &\text{ is a zero of } P'(x).\end{aligned}$$

- 4 Two roots of the equation $2x^3 + x^2 - px + 6 = 0$ are reciprocals of each other. What is the value of p ? 2

factors of 6 $\rightarrow 1, \pm 2, \pm 3, \pm 6$
Can't be 1 as reciprocal of 1 is 1.
try 2. reciprocal of 2 is $\frac{1}{2}$

$$\begin{aligned}2\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 - \frac{1}{2}p + 6 &= 0 \\ 2(8) + 4 - 2p + 6 &= 0 \\ 26 - 2p &= 0 \\ p &= 13 \checkmark\end{aligned}$$
$$\begin{aligned}2\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 - \frac{1}{2}p + 6 &= 0 \\ -\frac{1}{2}p + \frac{13}{2} &= 0 \\ \therefore p &= 13 \checkmark\end{aligned}$$

- 5 The polynomial $P(x) = x^3 - 4x^2 + tx + 10$ has roots $\alpha, -\alpha$ and β . Find the three roots and hence find the value of t . 2

$$\begin{aligned}\alpha + (-\alpha) + \beta &= 4 & \beta &= 4 \checkmark \\ \alpha(-\alpha)\beta &= -10 \\ -\alpha^2 \times 4 &= -10 \\ \alpha^2 &= \frac{5}{2} \\ \alpha &= \sqrt{\frac{5}{2}} \checkmark & \text{and } -\alpha &= -\sqrt{\frac{5}{2}} \\ \alpha\beta + \alpha(-\alpha) + (-\alpha)\beta &= t \\ -\alpha^2 &= t \\ \therefore t &= -\frac{5}{2}\end{aligned}$$

- 6 The polynomial equation $4x^3 - 12x^2 + 5x + 6 = 0$ has roots α, β and γ . It is known that one of the roots is the sum of the other two. Find α, β and γ . 2

$$\begin{aligned}
 &4x^3 - 12x^2 + 5x + 6 = 0 \\
 &\text{Let } \alpha, \beta \text{ and } \gamma \text{ be the roots of this equation} \\
 &\alpha + \beta + \gamma = 3 \quad (1) \quad \alpha = \beta + \gamma \quad (2) \\
 &\alpha\beta + \alpha\gamma + \beta\gamma = \frac{5}{4} \quad (3) \\
 &\alpha\beta\gamma = -\frac{3}{2} \quad (4) \\
 &\text{Sub (2) into (1)} \\
 &2\alpha = 3 \\
 &\alpha = \frac{3}{2} \\
 &\text{Sub } \alpha = \frac{3}{2} \text{ into (2)} \\
 &\frac{3}{2}\beta\gamma = -\frac{3}{2} \\
 &\beta\gamma = -1 \quad (5) \quad \text{and } \beta + \gamma = \frac{3}{2} \quad (6) \\
 &\text{From (6) } \beta = \frac{3}{2} - \gamma \\
 &\text{Sub } \beta \text{ into (5)} \\
 &(\frac{3}{2} - \gamma)\gamma = -1 \\
 &\frac{3}{2}\gamma - \gamma^2 = -1 \quad \therefore \alpha = \frac{3}{2} \\
 &\gamma^2 - \frac{3}{2}\gamma - 1 = 0 \quad \beta = 2 \\
 &2\gamma^2 - 3\gamma - 2 = 0 \quad \gamma = -\frac{1}{2} \\
 &(\gamma - 2)(2\gamma + 1) = 0 \\
 &\gamma = 2 \quad \gamma = -\frac{1}{2}
 \end{aligned}$$

- 7 The cubic equation $2x^3 - x^2 + x - 1 = 0$ has roots α, β and γ . Show without finding each individual root that:

a. $\alpha^2 + \beta^2 + \gamma^2 = -\frac{3}{4}$

$$\begin{aligned}
 &\alpha + \beta + \gamma = \frac{1}{2} \\
 &\alpha\beta + \alpha\gamma + \beta\gamma = \frac{1}{2} \\
 &\alpha\beta\gamma = \frac{1}{2} \\
 &\alpha^2 + \beta^2 + \gamma^2
 \end{aligned}$$

b. $\alpha^3 + \beta^3 + \gamma^3 = \frac{7}{8}$

$$\begin{aligned}
 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\
 &= (\frac{1}{2})^2 - 2 \times \frac{1}{2} = \frac{1}{4} - 1 = -\frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Sub } \alpha, \beta \text{ and } \gamma \text{ into } P(x) = 2x^3 - x^2 + x - 1 = 0 \\
 &2\alpha^3 - \alpha^2 + \alpha - 1 = 0 \quad (1) \\
 &2\beta^3 - \beta^2 + \beta - 1 = 0 \quad (2) \\
 &2\gamma^3 - \gamma^2 + \gamma - 1 = 0 \quad (3) \\
 &2(\alpha^3 + \beta^3 + \gamma^3) - (\alpha^2 + \beta^2 + \gamma^2) + (\alpha + \beta + \gamma) - 3 = 0 \\
 &2(\alpha^3 + \beta^3 + \gamma^3) + \frac{3}{4} + \frac{1}{2} - 3 = 0 \\
 &2(\alpha^3 + \beta^3 + \gamma^3) = \frac{7}{4} \\
 &\alpha^3 + \beta^3 + \gamma^3 = \frac{7}{8}
 \end{aligned}$$

- 8 Let $P(x) = x^2 + bx + c$ where b and c are constants. The zeros of $P(x)$ are α and $\alpha + 1$. What are the correct expressions for b and c in terms of α ? 2

$$\begin{aligned}
 &P(x) = x^2 + bx + c \quad (A) \\
 &\text{zeros are } x = \alpha \text{ and } x = \alpha + 1 \\
 &\text{sum of roots} = \alpha + \alpha + 1 \quad \text{product of roots} = \alpha(\alpha + 1) \\
 &= 2\alpha + 1 \quad = \alpha^2 + \alpha \\
 &\therefore 2\alpha + 1 = (-\frac{b}{a}) \quad \therefore \alpha^2 + \alpha = (\frac{c}{a}) \\
 &2\alpha + 1 = -\frac{b}{1} \quad \alpha^2 + \alpha = c \\
 &b = -(2\alpha + 1) \quad \therefore c = \alpha^2 + \alpha
 \end{aligned}$$

- 9 If two of the roots of $2x^3 + bx^2 + cx + 8 = 0$ are -1 and 2 , find the values of b and c .

2

$$\begin{aligned} (-1)(2)\alpha &= -4 \\ \alpha &= \frac{-4}{-2} = 2 \\ \therefore 2+2-1 &= \frac{-b}{2} & -2+2+4 &= \frac{c}{2} \\ b &= -6 & c &= 0 \\ b &= -6 \end{aligned}$$

- 10 Factorise and solve $2x^3 + x^2 - 7x - 6 = 0$.

2

$$\begin{aligned} \text{Let } P(x) &= 2x^3 + x^2 - 7x - 6 \\ P(-1) &= -2 + 1 + 7 - 6 = 0 \\ \therefore (x+1) &\text{ is a factor} \\ \begin{array}{r} 2x^2 - x - 6 \\ x+1 \overline{) 2x^3 + x^2 - 7x - 6} \\ \underline{2x^3 + 2x^2} \\ -x^2 - 7x \\ \underline{-x^2 - x} \\ -6x - 6 \\ \underline{-6x - 6} \\ 0 \end{array} & \therefore P(x) = (x+1)(2x^2 - x - 6) \\ & P(x) = (x+1)(2x+3)(x-2) \\ & \therefore x = -1, -\frac{3}{2}, 2 \end{aligned}$$

- 11 What is the coefficient of x when $P(x) = 6x^3 - 3x^2 - 2x - 1$ is divided by $(x - 1)$?

1

$$\begin{aligned} 6x^3 - 3x^2 - 2x - 1 &= (x-1)(6x^3 + 3x - 1) \\ \text{Coeff of } x &= 3 \end{aligned}$$

- 12 If α, β and γ are the roots of the equation $2x^3 + 3x^2 + 5x - 4 = 0$. Find the value of $\frac{1}{\alpha^2 + \beta^2 + \gamma^2}$.

2

$$\begin{aligned} \Rightarrow \alpha + \beta + \gamma &= -\frac{3}{2}, \quad \alpha\beta + \beta\gamma + \alpha\gamma = \frac{5}{2} \\ \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \\ &= \left(-\frac{3}{2}\right)^2 - 2\left(\frac{5}{2}\right) = \frac{9}{4} - \frac{10}{2} \\ &= -\frac{11}{4} \\ \frac{1}{\alpha^2 + \beta^2 + \gamma^2} &= -\frac{4}{11} \end{aligned}$$

- 13 The polynomial $2x^3 - 4x^2 + 3x - 6 = 0$ has roots α, β and γ . Calculate the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

2

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} \\ \alpha + \beta + \gamma &= \frac{-b}{a} = 2 \\ \alpha\beta + \beta\gamma + \gamma\alpha &= \frac{c}{a} = \frac{3}{2} \\ \alpha\beta\gamma &= -\frac{d}{a} = 3 \\ \therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{3/2}{3} = \frac{1}{2} \end{aligned}$$

14 The polynomial $P(x) = x^3 + bx^2 + cx + d$ has zeros 0, 3 and -3 .

a. Find b, c and d .

$$P(x) = x^3 + bx^2 + cx + d$$

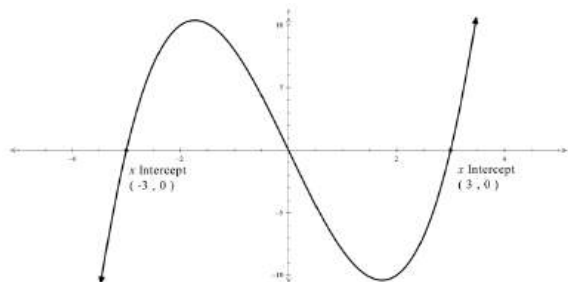
$$P(x) = x(x - 3)(x + 3) = x(x^2 - 9) = x^3 - 9x$$

$$b = 0 \quad c = -9 \quad d = 0$$

2

b. Without using calculus, sketch the graph of $y = P(x)$.

1



c. Hence, or otherwise, solve the inequality $\frac{x^2-9}{x} \geq 0$.

2

$$\frac{x^2 - 9}{x} > 0$$

$$\frac{x^2(x^2 - 9)}{x} > 0 \times x^2$$

$$x(x^2 - 9) > 0$$

From graph $-3 \leq x < 0$ or $x \geq 3$